MamNet-6: Multi-View Fusion, CrossView Transformer, and Uncertainty-Aware Triage for Breast Cancer Detection on VinDr-Mammo

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***Abstract*—Most published work on the VinDr-Mammo full-field digital mammography (FFDM) benchmark looks at one view at a time, although radiologists read the craniocaudal (CC) and mediolateral oblique (MLO) views together. To test whether sharing information between the two views improves breast-level cancer detection, we built MamNet-6, a six-stage pipeline that adds multi-view fusion, spatial cross-attention, uncertainty estimation and a hybrid triage simulation, and tested it zero-shot on the RSNA 2022 dataset. The best of five single-view convolutional neural networks (CNNs), DenseNet-121, reached an area under the receiver operating characteristic curve (AUROC) of 0.756. BreastPairNet, which passes both views through a shared backbone, reached 0.826 (95% confidence interval [CI] 0.776–0.871), and the CrossView Transformer reached 0.832 (0.783–0.875). These stage-to-stage differences have overlapping CIs and are not individually established, and among configurations sharing a backbone, CrossView had lower observed AUROC than pooled fusion. A 16-model ensemble with test-time augmentation reached 0.846, with a sensitivity of 0.657 at 90% specificity and a positive predictive value (PPV) of 0.184. Letting the AI read 50% of exams kept 87.9% sensitivity with an oracle reader for the rest. Bounding boxes drawn on training images lowered AUROC by 0.12–0.20, which suggests shortcut learning, and zero-shot transfer to RSNA 2022 gave an AUROC of 0.601, a large gap between the two datasets.**

***Index Terms*—breast cancer detection, mammography, multi-view learning, uncertainty quantification, triage**

I. Introduction

BREAST cancer is the most frequently diagnosed cancer and the leading cause of cancer death among women worldwide, accounting for 11.8% of all new cancer cases across both sexes in 2024 [19]. Screening with full-field digital mammography (FFDM) lowers breast-cancer mortality but takes a great deal of radiologist time, and earlier hand-engineered detection systems brought no measurable benefit in routine screening [16]. Convolutional neural networks (CNNs) now match or exceed radiologists in area under the receiver operating characteristic curve (AUROC) [10], and meta-analyses report a pooled AUROC of 0.87 for AI on digital mammography against 0.81 for radiologists [4].

Three lines of work motivate this study, each leaving a gap on the four-view VinDr-Mammo benchmark [1], which carries Breast Imaging Reporting and Data System (BI-RADS) labels. First, models that fuse the craniocaudal (CC) and mediolateral oblique (MLO) projections improve over single-view inference because findings often appear unambiguously in only one view [9], yet most published VinDr-Mammo results still use one image at a time. Second, transformer-based fusion lets views interact at the feature level, but most designs simply concatenate global features. Third, Monte Carlo (MC) Dropout uncertainty [12] enables *hybrid triage*, in which the AI reads the exams it is most confident about and refers the rest; Verboom et al. [11] cut workload by 38.1% on 41,469 Dutch screening exams while preserving the cancer detection rate. No published study has applied MC Dropout to VinDr-Mammo and simulated such a policy. On VinDr-Mammo, MamT4 [17] reports AUROC=0.840±0.017; other work includes the contrast-limited adaptive histogram equalisation (CLAHE) and YOLOv12 gains in mean average precision shown by Abdikenov et al. [2], and the competitive patch-level F1 under limited annotation reported by Mellado et al. [3].

Our central question is whether sharing information between the CC and MLO views improves breast-level detection over single-view CNNs (Stage 1). We test two ways: fusing pooled per-view features (Stage 2, BreastPairNet) and spatial cross-attention before pooling (Stage 3, CrossView Transformer). MamNet-6 then adds MC Dropout uncertainty (Stage 4), an ensemble with test-time augmentation (Stage 5), and hybrid triage with external validation on RSNA 2022 (Stage 6). All stages share preprocessing, Stage 3b is an ablation of Stage 3, and Stage 5 ensembles Stage 2 models. We present explanations of results as hypotheses.

II. Datasets

A. VinDr-Mammo (primary dataset)

VinDr-Mammo [1] is a Vietnamese FFDM benchmark of 20,000 Digital Imaging and Communications in Medicine (DICOM) images from 5,000 patients (R-CC, L-CC, R-MLO, L-MLO per study), with BI-RADS (1–5), density (A–D) and lesion bounding-box annotations. BI-RADS 1–3 images (14,036, 4,144 and 832) are labelled negative and BI-RADS 4–5 (824 and 164) positive, giving 988 positive images (4.94%).

The official split is 4,000 training patients (16,000 images) and 1,000 test patients (4,000 images). Taking one ipsilateral CC+MLO pair per breast gives 8,000 training and 2,000 test pairs, of which 99 test pairs (4.95%) are cancer-positive. We

hold out a 5% validation slice (about 400 pairs) for early stopping, and the official test split is never used for model selection.

B. RSNA 2022 (external dataset)

The Radiological Society of North America (RSNA) 2022 screening dataset contains 54,706 screening DICOMs from 11,913 patients, normalised as for VinDr-Mammo. Excluding non-CC/MLO views leaves 23,826 CC+MLO pairs (19,062 training, 4,764 validation; 80/20 patient-level split).

III. Methods

A. DICOM Preprocessing

Each image is decoded with pydicom and `pylibjpeg`, clipped to its 1st–99th intensity percentiles, enhanced with CLAHE [13] (32×32 tiles, clip limit 2.0), and flipped so the breast faces right whenever the left half is brighter (breast tissue is brighter than background). Output is an 8-bit PNG replicated to three channels.

B. Single-view CNN Backbones (Stage 1)

We compare five configurations: VGG19, ResNet-50 [6] at 224 and 384 px, and EfficientNet-B3 [8] and DenseNet-121 [7] at 224 px. All use ImageNet weights, focal loss [5] (α=0.25, γ=2.0; an algorithm-level imbalance method [14]), a WeightedRandomSampler and AdamW with cosine decay, trained for 20 epochs with early stopping on validation AUROC.

C. BreastPairNet (Stage 2)

A shared backbone turns each view into a global-average-pooled vector, $\mathbf{f}_{CC}, \mathbf{f}_{MLO} \in \mathbb{R}^d$ (Fig. 1(a)). The two vectors are fused either by concatenation, followed by a $2d$-dimensional multilayer perceptron (MLP) with dropout 0.5, or by attention with learned per-view scalar weights (softmax). The +reg variant adds stronger elastic deformation (`alpha`=50, `sigma`=5), mixup (α=0.2) and label smoothing (0.1). The backbone is frozen for 3 epochs, then all parameters train at learning rate 1e-4.

D. CrossView Transformer (Stage 3)

Spatial cross-attention replaces pooled fusion (Fig. 1(b)). The last spatial feature maps of the shared backbone (DenseNet-121, EfficientNet-B3 or ConvNeXt-Base [15]) are flattened into tokens and passed through a two-layer Transformer encoder in which each view's queries attend to the other view's keys and values. The outputs are then global-averaged and concatenated for the classifier.

E. Bounding-box Annotation (Stage 3b)

Ground-truth boxes from VinDr-Mammo were drawn as white outlines, 40 px wide, on training images only. For RSNA 2022, a YOLOv8s detector pre-trained on VinDr-Mammo supplied boxes (0.8% sensitivity; noisy hints only). Test images remained unannotated to prevent leakage from the box annotations.

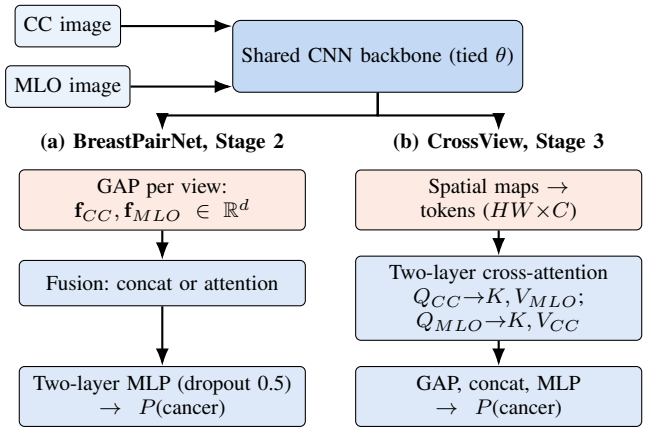


Fig. 1. Two alternative multi-view architectures, trained and evaluated separately; in each, CC and MLO images share one CNN backbone (tied θ). (a) BreastPairNet global-average-pools (GAP) each view and fuses the pooled vectors. (b) The CrossView Transformer instead applies two-layer bidirectional cross-attention to spatial tokens *before* pooling. Orange marks the representation passed to fusion: pooled vectors in (a), spatial tokens in (b).

F. MC Dropout (Stage 4)

MC Dropout [12] keeps dropout on at test time (BatchNorm in eval mode), giving a predictive distribution without retraining. We run $N = 20$ stochastic forward passes per test pair, classify with the mean probability $\bar{p}$ of the samples $\{p_i\}_{i=1}^{20}$, and measure uncertainty with the binary entropy $H(\bar{p}) = -\bar{p}\log\bar{p} - (1-\bar{p})\log(1-\bar{p})$. The classification threshold maximises Youden's J index [18] (J = sensitivity + specificity -1) on the validation set. Stage 1 baselines (Table I) use the default 0.5 threshold.

G. Ensemble (Stage 5)

The ensemble averages the predicted probabilities of its member checkpoints, each of which is itself averaged over five test-time augmentation (TTA) passes. The primary ensemble combines 16 BreastPairNet checkpoints with three breast-asymmetry models, which score each breast together with the opposite breast of the same patient. Section IV-F compares two variants.

H. Hybrid Triage (Stage 6)

Following Verboom et al. [11], we sort cases by predictive entropy and send the most confident fraction K to the AI and the rest to a radiologist. AI decisions come from the MC Dropout DenseNet-121/concat model (Stage 4), which calls a breast positive above its Youden-optimal threshold (0.3138). Overall sensitivity is $(\mathrm{TP}_{\mathrm{AI}} + s_{\mathrm{reader}}\, N_{\mathrm{pos,referred}})/N_{\mathrm{pos}}$, with reader sensitivity $s_{\mathrm{reader}} = 1$ for an oracle reader or 0.85.

I. Training and evaluation

All models were trained on one NVIDIA RTX 5070 (12 GB) with FP16 automatic mixed precision and AdamW (weight decay 1e-4 for Stage 1 and 1e-3 for Stage 2). AUROC is the primary metric. We also report F1 at the Youden threshold, sensitivity, specificity, S@90%spec and S@95% (sensitivity

TABLE I
SINGLE-VIEW CNN BASELINES ON VINDR-MAMMO (N=16,000 TRAIN, N=4,000 TEST IMAGES). SENSITIVITY, SPECIFICITY AND F1 AT THE DEFAULT 0.5 THRESHOLD.

Architecture	Image size	Sensitivity	Specificity	F1	AUROC	S@90%
VGG19	224 px	0.2626	0.9835	0.3323	0.7066	0.3990
ResNet-50	224 px	0.2677	0.9847	0.3430	0.7286	0.4495
EfficientNet-B3	224 px	0.2980	0.9687	0.3138	0.7330	0.3990
ResNet-50 (384 px)	384 px	0.2879	0.9821	0.3529	0.7489	0.4495
DenseNet-121	**224 px**	**0.2475**	**0.9882**	**0.3356**	**0.7564**	**0.4444**

at 90% and 95% specificity), and PPV, the positive predictive value TP/(TP+FP), where TP and FP are the true and false positives at the Youden-optimal threshold. 95% percentile bootstrap confidence intervals (CIs; n=2000 resamples) are computed on the 2,000-pair test split (99 cancer-positive). No paired significance tests between models were performed, so no difference between models is reported as statistically significant.

IV. RESULTS AND DISCUSSION

A. Stage 1: Single-view baselines

DenseNet-121 has the best AUROC (0.7564) but only 24.8% sensitivity at the default threshold (Table I). All five models overfit (train F1 $\approx$ 0.98), and at 90% specificity they reach only 40–45% sensitivity, well below the roughly 85% sensitivity of radiologists in screening [16].

B. Stage 2: Multi-view BreastPairNet

The best model, EfficientNet-B3 with attention fusion and +reg, reaches AUROC=0.8256 (95% CI 0.7755–0.8707), a 9.1% relative gain over single-view DenseNet-121 (Table II). This is consistent with lesions appearing unambiguously in only one projection, and conditioning on both views may reduce view-specific false positives, though shared weights and fusion cannot be separated. The comparison also changes from image-level to breast-level scoring, so it does not isolate the effect of multi-view fusion. Because each configuration was trained once, the CIs in Table II and Table III reflect test sampling, not run-to-run variation. The best Stage 2 and Stage 3 models peaked at epochs 7 and 8, then overfit.

The +reg recipe raised AUROC by 0.008 to 0.021 for every backbone and fusion pair. With only about 395 cancer-positive training pairs, the extra augmentation probably acts as a regulariser. Higher resolution did not always help: DenseNet-121 gained 0.014 at 512 px, but EfficientNet-B3 with +reg at 224 px (0.8256) beat the 512 px version without +reg (0.8175). ResNet-50 got worse at 384 px (0.7467 vs 0.7962), possibly because the batch had to shrink to fit GPU memory.

C. Stage 3: CrossView Transformer

ConvNeXt-Base gives the best CrossView result, AUROC=0.8317 (10.0% above single-view DenseNet-121). That is only 0.006 above the best BreastPairNet, and we did not test the difference statistically. Among reported configurations sharing a backbone, CrossView had lower observed AUROC than BreastPairNet (DenseNet-121 0.7874 vs 0.8073; EfficientNet-B3 0.7795 vs 0.8048; unpaired single runs). ConvNeXt-Base's result may therefore owe more to the backbone than to cross-attention, and the benefit of preserving spatial information before pooling remains a hypothesis. Joint VinDr-Mammo+RSNA 2022 training gives 0.8028 with DenseNet-121, suggesting a modest cross-dataset benefit (single runs, not tested). Chest-X-ray pretraining from torchxrayvision[1] does worse than ImageNet initialisation (0.7648 vs 0.7874). Domain mismatch between lung radiographs and mammograms may explain this, tentatively, as the configurations also differ in other ways.

D. Stage 3b: Bounding-box annotation ablation

Drawing boxes on the training images lowered AUROC by 0.122–0.201 in all four pairs (Table IV). Training F1 climbs to about 0.97 once the backbone unfreezes at epoch 4, while test AUROC falls. We hypothesise that the network learns the white outlines as a texture shortcut that tracks the label in training but is missing at test time, consistent with shortcut learning in medical imaging; we did not test this directly. Box-dropout, test-time box injection from a lesion detector, or cropping to the box region might help but are untested.

E. Stage 4: MC Dropout uncertainty quantification

MC Dropout raises AUROC only slightly (+0.003, within the bootstrap CI). Sensitivity rises from 0.7374 to 0.7677 while specificity falls from 0.7249 to 0.7007, each at its own Youden threshold (Table V). More useful is the entropy, with mean 0.5670 on cancer-positive pairs and 0.5112 on negatives (Table V); the Stage 6 triage policy relies on this 0.056-nat gap.

F. Stage 5: Model ensemble

The primary ensemble (16 BreastPairNet checkpoints with three breast-asymmetry models and TTA$\times$5; Section III-G) gives the best result in the study, AUROC=0.8458. At threshold 0.3414 it has TP=73, TN=1577, FP=324 and FN=26 (other metrics in Table VI). It does not include ConvNeXt-Base or EfficientNet-B3 +reg models. Single-model PPV ranges from 0.109 to 0.263 (Stage 2) and 0.117 to 0.169 (Stage 3), in line with the 4.95% positive prevalence. Screening prevalence is usually lower, so deployed PPV would be lower; clinical use would require calibrated predictions and prevalence-adjusted thresholds.

One variant adds all CrossView checkpoints including lower-epoch snapshots (25 models); the other is a selective ensemble of the top-3 +reg BreastPairNet, top-5 CrossView and three asymmetry models (11 models). Both are within 0.0017 AUROC of the primary ensemble; with no ensemble CIs reported and no paired test, these differences are unassessed. Adding the CrossView checkpoints, ConvNeXt-Base among them, did not raise AUROC; as several were

[1]torchxrayvision: `github.com/mlmed/torchxrayvision`

TABLE II
BREASTPAIRNET ABLATION ACROSS BACKBONE, FUSION, AND RESOLUTION. SENS: SENSITIVITY.

Architecture	Fusion	Image	Best AUROC [95% CI]	F1	Sens	PPV	S@90% [95% CI]
ResNet-50	concat	384 px	0.7467 [0.6921–0.7998]	0.2350	0.5354	0.151	0.4646 [0.3736–0.5673]
ResNet-50	attention	224 px	0.7847 [0.7324–0.8338]	0.2522	0.5859	0.161	0.5051 [0.4023–0.6019]
ResNet-50	attention	512 px	0.7922 [0.7339–0.8429]	0.2093	0.7475	0.122	0.5152 [0.4130–0.6214]
ResNet-50	concat	224 px	0.7962 [0.7428–0.8445]	0.2363	0.6970	0.142	0.4949 [0.3958–0.5977]
DenseNet-121	attention	224 px	0.8014 [0.7487–0.8515]	0.2800	0.6364	0.179	0.5758 [0.4660–0.6667]
EfficientNet-B3	attention	224 px	0.8048 [0.7530–0.8556]	0.2138	0.7677	0.124	0.5455 [0.4476–0.6449]
DenseNet-121	concat	224 px	0.8073 [0.7577–0.8504]	0.2101	0.7374	0.122	0.5152 [0.4141–0.6100]
DenseNet-121 **+reg**	attention	224 px	0.8097 [0.7594–0.8591]	0.2033	0.7980	0.117	0.5556 [0.4301–0.6458]
DenseNet-121	attention	512 px	0.8150 [0.7675–0.8617]	0.1928	0.8081	0.109	0.5455 [0.4476–0.6408]
DenseNet-121 **+reg**	concat	224 px	0.8177 [0.7659–0.8649]	0.2357	0.7677	0.139	0.6061 [0.5047–0.7000]
EfficientNet-B3	attention	512 px	0.8175 [0.7698–0.8661]	0.2807	0.7273	0.160	0.5556 [0.4495–0.6505]
EfficientNet-B3 +reg	**attention**	**224 px**	**0.8256 [0.7755–0.8707]**	**0.3701**	**0.6263**	**0.263**	**0.6263 [0.5229–0.7143]**

PPV = TP/(TP+FP) at the Youden-optimal threshold. At 4.95% prevalence, PPV is inherently low; higher PPV reflects a more conservative threshold (lower sensitivity). All CIs via 2,000-resample percentile bootstrap on the 2,000-pair test split.

TABLE III
CROSSVIEW TRANSFORMER RESULTS WITHOUT BOUNDING-BOX ANNOTATION. SENS: SENSITIVITY.

Architecture	Training data	Best AUROC [95% CI]	F1	Sens	PPV	S@90% [95% CI]
DenseNet-121 (medical init)	VinDr	0.7648 [0.7116–0.8210]	0.2374	0.5960	0.148	0.5051 [0.4020–0.6100]
EfficientNet-B3	VinDr	0.7795 [0.7276–0.8300]	0.2022	0.7273	0.117	0.4545 [0.3474–0.5500]
DenseNet-121	VinDr	0.7874 [0.7299–0.8401]	0.2678	0.6465	0.169	0.5556 [0.4474–0.6482]
DenseNet-121	VinDr+RSNA	0.8028 [0.7484–0.8536]	0.2640	0.6667	0.165	0.5354 [0.4433–0.6404]
ConvNeXt-Base	**VinDr**	**0.8317 [0.7832–0.8753]**	**0.2587**	**0.7475**	**0.156**	**0.5758 [0.4737–0.6768]**

TABLE IV
BOUNDING-BOX ANNOTATION ABLATION. TEST SET ALWAYS USES ORIGINAL UNANNOTATED PNGS; ΔAUROC IS RELATIVE TO THE UNANNOTATED MODEL IN TABLE III.

Architecture	With annotation AUROC [95% CI]	ΔAUROC
DenseNet-121	0.5862 [0.5274–0.6433]	-0.201
DenseNet-121 +RSNA	0.6675	-0.135
DenseNet-121 (medical)	0.6433 [0.5886–0.6966]	-0.122
EfficientNet-B3	0.6020 [0.5460–0.6617]	-0.178

TABLE V
MC DROPOUT VS DETERMINISTIC INFERENCE (DENSENET-121/CONCAT), AND PREDICTIVE ENTROPY BY CLASS ($N = 20$ MC PASSES, 2,000 TEST PAIRS). SENS/SPEC: SENSITIVITY/SPECIFICITY; THR.: THRESHOLD.

Mode	AUROC	Sens	Spec	S@90%	Thr.
Deterministic	0.8073	0.7374	0.7249	0.5152	0.3304
MC Dropout	**0.8099**	**0.7677**	**0.7007**	**0.5253**	**0.3138**

Entropy	Negatives ($n = 1{,}901$)	Positives ($n = 99$)
Mean	0.5112	0.5670
Median	0.5310	0.6221
Std dev	0.1436	0.1370

Gap: +0.056 nats. Higher entropy ⇒ refer high-uncertainty cases.

added at once, ConvNeXt-Base's own effect is not isolated. We have two unmeasured hypotheses. Ensembles gain from error diversity rather than from the rank of their members, and the CrossView models may make correlated errors. Also,

TABLE VI
ENSEMBLE VARIANTS ON VINDR-MAMMO (2,000 PAIRS, 99 CANCERS) AND THE PRIMARY ENSEMBLE ZERO-SHOT ON RSNA 2022 (4,764 PAIRS, 110 CANCERS); TTA×5, MEAN PROBABILITY, OPERATING-POINT METRICS AT EACH YOUDEN-OPTIMAL THRESHOLD.

Ensemble	N	AUROC	Sens	Spec	PPV	F1	S@90%	S@95%
Primary	16	0.8458	0.7374	0.8296	0.184	0.2944	0.6566	0.5657
+ CrossView	25	0.8441	0.7677	—	—	—	0.6566	—
Selective	11	0.8446	—	—	—	—	0.6667	—
RSNA[c]	16	0.6008	0.3909	0.7598	—	—	0.1636	—

Sens/Spec: sensitivity/specificity. —: not reported. [c]AUROC 95% CI 0.5577–0.6585. PPV=0.184 means 1 in 5.4 recalls is cancer-positive at this threshold; at 4.95% test prevalence, this is consistent with reported breast-cancer AI triage literature (PPV 15–25%).

AUROC depends only on how averaged scores rank cases, so members on a different probability scale can dominate the average and change that ranking even if each ranks cases well alone. Measuring error correlation and calibrating members would test this.

G. Stage 6: Hybrid triage simulation

With an oracle reader, AI reading of K=50% of exams keeps 87.9% overall sensitivity and halves radiologist workload (Table VII); with a realistic 85% radiologist sensitivity [16], it gives 77.3%, below the clinical floor. Keeping at least 80% overall sensitivity under realistic conditions requires K≤40%.

TABLE VII
HYBRID TRIAGE SWEEP. FRAC: AI-READ SHARE; WKLD Δ: RADIOLOGIST WORKLOAD CHANGE; SEN (OR.)/(85%R): OVERALL SENSITIVITY WITH AN ORACLE/85%-SENSITIVE READER; AI SEN: SENSITIVITY ON AI-READ EXAMS; RECALL: AI RECALL RATE.

Frac	Wkld Δ	Sen (or.)	Sen (85%r)	AI Sen	Recall
30%	-30%	0.9495	0.8253	0.7059	0.0017
40%	-40%	0.9192	0.8071	0.6800	0.0013
50%	**-50%**	**0.8788**	**0.7727**	**0.5862**	**0.0031**
60%	-60%	0.8283	0.7359	0.5526	0.0052
70%	-70%	0.7677	0.6935	0.5400	0.0133
80%	-80%	0.7677	0.7207	0.6618	0.1305
90%	-90%	0.7677	0.7450	0.7262	0.2238

TABLE VIII
BEST MODEL PER STAGE ON THE VINDR-MAMMO TEST SPLIT. Δ: OBSERVED CHANGE FROM THE PREVIOUS ROW (UNPAIRED, UNTESTED). STAGES 4 AND 6 ADD NO NEW CLASSIFIER.

Stage	Model	AUROC [95% CI]	Δ	Caveat
1	DenseNet-121	0.7564	—	image-level
2	EffB3 +reg	0.8256 [0.776–0.871]	+0.069	breast-level
3	ConvNeXt-Base	0.8317 [0.783–0.875]	+0.006	new backbone
5	Ensemble (16)	0.8458	+0.014	no CI reported

H. External validation and comparison to literature

Applied zero-shot to RSNA 2022, the best ensemble drops sharply (Table VI). The datasets may differ in population (e.g., breast density, lesion prevalence) [1], [10] and in FFDM equipment and post-processing; the 75.1% drop in S@90% suggests the conservative end of the decision surface suffers most. Our best ensemble scores higher than MamT4 [17] (AUROC=0.840), but MamT4 excludes BI-RADS 3 cases from evaluation while our pipeline keeps them as negatives, so the two values are not directly comparable. McKinney et al. [10] (AUROC=0.889) and Yoon et al. [4] (pooled AUROC=0.87) likewise use different data. Beyond AUROC, MamNet-6 ablates multi-view fusion, regularisation, spatial cross-attention and ensemble size, which MamT4 does not systematically ablate, and adds uncertainty estimation and triage simulation based on a published framework [11].

I. Overall contribution and statistical caveats

Table VIII lists the best model at each stage; Stage 1 is image-level while later stages are breast-level, and the best Stage 2 and Stage 3 models use different backbones. With 99 cancer-positive test pairs, individual 95% CIs span ±0.04–0.05 AUROC. Overlapping CIs do not settle a comparison on the same cases, which needs a paired test we did not perform, so the 9.1%, 10.0% and 11.8% gains and the ablation are observed results, not established improvements.

Limitations. Only 99 positive test pairs limit fine-grained comparison, the domain gap between VinDr-Mammo and RSNA 2022 is large, and 7,054 non-CC/MLO RSNA images were excluded from pairing. Each configuration was trained once (one seed, no cross-validation), so run-to-run variance is unknown, and no paired significance tests were performed.

Future work. Priorities are repeated runs or five-fold cross-validation, keeping per-case predictions to allow paired tests, a same-backbone comparison to isolate cross-attention, and box-dropout to test the shortcut hypothesis. CBIS-DDSM/INbreast validation, RSNA 2022 fine-tuning and a prospective triage study would test generalisation and the assumed 85% reader.

V. CONCLUSION

MamNet-6 raises VinDr-Mammo AUROC from 0.7564 (best single-view CNN) to 0.8458 (16-model ensemble), an observed but untested 11.8% gain. Training-image boxes lowered AUROC by 0.12–0.20, plausibly through shortcut learning, and among configurations sharing a backbone, CrossView had lower observed AUROC than BreastPairNet. With MC Dropout triage, 50% AI reads keep 87.9% sensitivity with an oracle reader but 77.3% with an 85%-sensitive reader, so K≤40% is needed to stay above the 80% floor. RSNA 2022 (AUROC=0.6008) shows a generalisation gap.

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